
Software Requirements Specification

For Senior Citizen Digital Services App

Version: 1.0

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

1.1 Document Purpose

This document describes the software requirements for Sahaaya – Senior Citizen Digital Services App. It defines the system features, scope, and requirements needed for development, testing, and evaluation of the application.

The SRS provides a clear description of system behavior, functional requirements, and constraints so that developers, testers, and stakeholders understand how the system should work.

1.2 Product Scope

Sahaaya is a mobile application designed to help senior citizens access important digital services easily. The application provides features such as medicine reminders, emergency alerts, healthcare information, and daily routine reminders.

The main objective of the system is to simplify technology for elderly users by providing a simple interface with large icons, voice support, and easy navigation.

1.3 Intended Audience and Document Overview

This document is intended for developers, project team members, testers, and the course instructor. It explains the system overview, requirements, use cases, and design details for Sahaaya application. Readers should begin with the introduction and overview sections to understand the system's context, then review the functional requirements and use cases to understand system behavior.

1.4 Definitions, Acronyms and Abbreviations

(API) – Application Programming Interface
(Firebase) – Cloud platform used for database and authentication
(GPS) – Global Positioning System used for location services
(SOS) – Emergency alert feature used to contact caregivers or family
(SRS) – Software Requirements Specification document
(UI) – User Interface of the application

1.5 Document Conventions

This document follows the IEEE SRS format. The text uses Arial font with numbered headings and subheadings. Functional requirements are labeled systematically for clarity and easy reference.

1.6 References and Acknowledgments

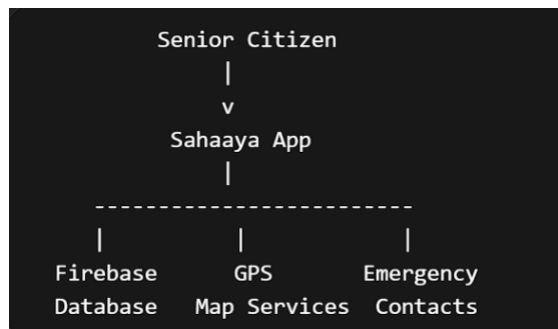
References include the IEEE SRS template, Software Engineering course materials, the project Statement of Work, and documentation related to Android development, Firebase services, and mobile accessibility design.

2 Overall Description

2.1 Product Overview

Sahaaya is a mobile application created to support senior citizens by making the digital services easier to access every day. Many elderly people find modern apps complicated, so this system focuses on simplicity and accessibility.

The application brings together useful features such as health reminders, emergency assistance, and important service information in one place. It runs on a smartphone and connects with services like Firebase for storing data, GPS for location features, and notification systems for alerts. The goal is to provide a helpful and easy-to-use digital assistant for senior citizens.



2.2 Product Functionality

The main functions of the Sahaaya mobile application includes:

- Creating and managing user accounts
- Setting reminders for medicines
- Scheduling doctor appointment reminders
- Providing daily routine alerts
- Sending emergency SOS alerts to contacts
- Storing health records and prescriptions
- Showing nearby hospitals using location services
- Providing voice assistance and text-to-speech support
- Offering access to government services for seniors

2.3 Design and Implementation Constraints

The application will be developed as an Android mobile app using tools such as Android Studio and Firebase for database and authentication services. Some features may also rely on backend services built using Node.js.

The system design will follow the COMET method to structure the software architecture. UML diagrams will also be used to represent system design, use cases, and interactions between different components. The project must also be completed within the academic timeline and available resources.

2.4 Assumptions and Dependencies

The design of this system assumes that users have access to a smartphone and basic internet connectivity. It is also assumed that senior citizens will be able to interact with simple interfaces that use large icons and clear navigation.

The application depends on external services such as Firebase for storing user data and GPS for location-based features. Emergency contacts must also be set up by the user or caregiver for the SOS feature to work properly.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The Sahaaya application is designed with a simple and easy-to-use interface so that senior citizens can interact with it comfortably. The app will run on a smartphone, and users will interact with it using the touchscreen.

The main screen shows clear options such as voice chat, medicine reminders, emergency SOS, health records, and nearby hospitals. Large buttons, readable text, and simple navigation help users move through the app without confusion. Voice assistance and text-to-speech may help users who find reading difficult.

SAHAAYA APP
[Voice Chat]

[Medicine Reminder]

[Doctor Appointment]

[Emergency SOS]

[Health Records]

[Nearby Hospitals]

Users simply tap the buttons on the screen to access different features. Or give a voice command for the same.

3.1.2 Hardware Interfaces

The system interacts with the hardware components of the user's smartphone.

- Touchscreen – Allows users to select options and interact with the application.
- GPS sensor – Helps the app identify the user's location and display nearby hospitals or emergency services.
- Device storage – Stores application data and health records.
- Speaker – Used for voice assistance and reading information aloud.
- Internet connection – Allows the application to communicate with cloud services such as Firebase.

These hardware components allow the application to provide reminders, emergency alerts, and location-based services.

3.1.3 Software Interfaces

The Sahaaya application connects with different software services to perform its functions.

The system uses Firebase for user authentication and storing user information such as reminders and health records. Notification services are used to send alerts for medicine reminders and appointments. The application also uses map or GPS services to show nearby hospitals or important locations.

These software services work together to support the main functions of the application.

3.2 Functional Requirements

F1. The system shall allow users to create an account and register their personal information.

F2. The system shall allow registered users to log in securely.

F3. The system shall provide voice assistance and text-to-speech support for better accessibility.

F4. The system shall notify users about scheduled doctor appointments.

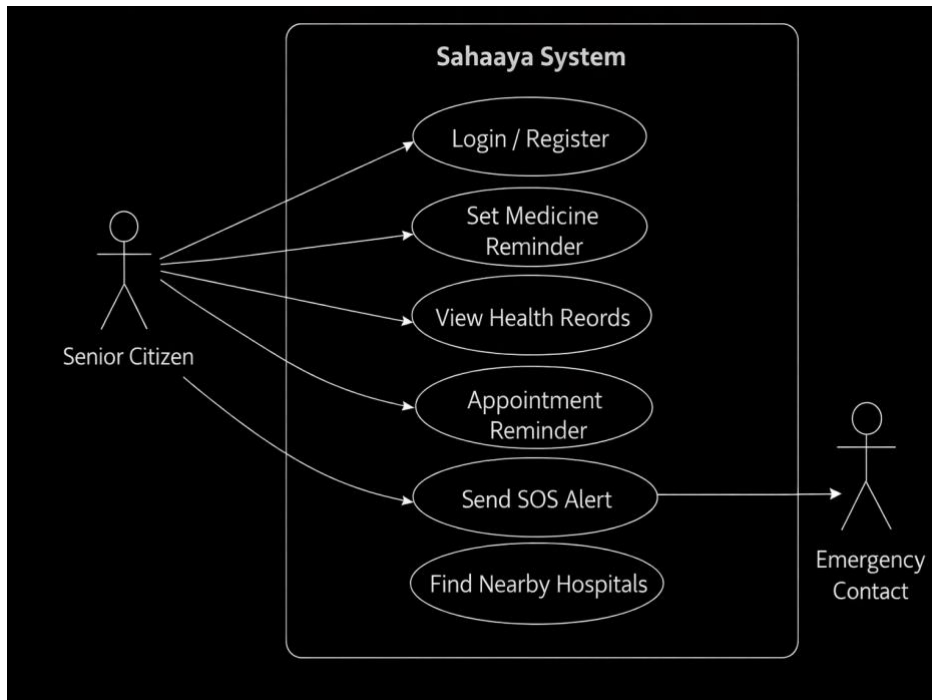
F5. The system shall provide an emergency SOS feature that alerts emergency contacts.

F6. The system shall allow users to store and view health records and prescriptions.

F7. The system shall display nearby hospitals using location services.

F8. The system shall allow users to set reminders for taking medicines.

3.3 Use Case Model



3.3.1 Use Case U1 – Set Medicine Reminder

Author: Sahithi

Purpose: This use case allows users to set reminders, so they get reminded to take their medicines on time.

Requirements Traceability: F3 – Medicine Reminder

Priority: High

Preconditions: The user must be logged into the application.

Postconditions: The reminder is saved and the system will notify the user at the selected time.

Actors: Senior Citizen

Extends: None

Flow of Events

Basic Flow

1. The user opens the reminder section.
2. The user enters the medicine name and dosage.
3. The user sets the reminder time.
4. The system saves the reminder.
5. The system sends a notification at the scheduled time.

Alternative Flow: The user may use the voice feature to tell the reminder to the Sahaaya app.

Exceptions: If notifications are turned off on the device, the reminder alert may not appear.

Includes: None

Notes: Reminder alerts depend on the device notification settings.

3.3.2 Use Case U2 – Emergency SOS

Author: Sahithi

Purpose: This use case allows users to quickly send an emergency alert to their saved contacts.

Requirements Traceability: F5 – Emergency SOS

Priority: High

Preconditions: Emergency contacts must already be added to the system.

Postconditions: Emergency contacts receive a message or alert with the user's location.

Actors: Senior Citizen, Emergency Contact

Basic Flow

1. The user presses the SOS button.
2. The system sends an alert to emergency contacts.
3. The user's location is shared with the contacts.

Alternative Flow: The user may use the voice feature which automatically sends alert to caregiver and user's selected contacts.

4 Other Non-functional Requirements

4.1 Performance Requirements

P1. The system should load the main application screen within 5 seconds after the user opens the app.

P2. Medicine reminders and appointment alerts should be delivered at the scheduled time without noticeable delay.

P3. When the SOS button is pressed, the emergency alert should be sent to the saved contacts within 5 seconds.

P4. The system should be able to store and retrieve health records quickly without affecting the overall app performance.

These requirements ensure that the application works smoothly and responds quickly when users interact with important features such as reminders or emergency alerts.

4.2 Safety and Security Requirements

The system should ensure that users' personal and medical information is kept safe and private.

S1. The application should require user authentication (login or registration) before accessing personal information.

S2. User data such as health records and reminders should be stored securely in the database.

S3. The system should only allow authorized users to access their own data.

S4. Emergency contact information should be protected and only used when the SOS feature is activated.

S5. Communication between the mobile application and cloud services should use secure connections to prevent unauthorized access.

These measures help protect user data and ensure that the system is safe to use.

4.3 Software Quality Attributes

The application should function consistently without crashes or major errors. Important features such as reminders and emergency alerts must work reliably so that users can depend on the system in daily situations.

The system should also be able to handle multiple reminders and notifications without affecting performance.

4.3.1 Usability

Since the application is designed for senior citizens, it must be easy to use and understand.

The interface should include large buttons, clear text, simple navigation, and voice assistance to help elderly users interact with the system comfortably.

4.3.2 Maintainability

The system should be designed in a modular way so that developers can easily update or improve individual features without affecting the entire application.

This also allows new services or features to be added in future versions of the system.

5 Other Requirements

The application should support multiple languages (English, Hindi, Telugu, Tamil at the beginning) so that users from different regions can easily understand the interface. The system should also be designed so that additional services, such as new healthcare features or government services, can be integrated in future updates.

Appendix A – Data Dictionary

The data dictionary contains the main data elements used in the Sahaaya system. It helps describe the different inputs, outputs, and variables used by the application. This makes it easier for developers to understand how data is stored and used in the system.

Data Item	Description	Type
User ID	Unique identifier for each registered user	Integer
User Name	Name of the user using the application	Text
Medicine Name	Name of the medicine added to reminders	Text
Reminder Time	Time set for medicine or appointment reminder	Time
Emergency Contact	Phone number of the saved emergency contact	Number
Health Record	Stored medical record or prescription image	File / Text
Location Data	User location used to find nearby hospitals	GPS Data

Appendix B - Group Log

This section records the activities and meetings conducted by the project team during the development of the Sahaaya system. It helps track the progress of the project and the contributions made by each team member.

The team discussed project ideas, divided responsibilities, and reviewed progress during regular meetings. Tasks such as requirement analysis, system design, development planning, and documentation preparation were completed collaboratively.

Date	Activity
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Meeting 1	Discussed project idea and came up with the idea "Sahaaya"
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Meeting 2	Designed use case diagram for Sahaaya
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Meeting 3	Designed IBM architecture
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Meeting 4	Prepared SRS document
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Meeting 5	Prepared a prototype for the app interface
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